No. of Functional Features Included in the Solution

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Comprehensive Matrix of Functional Features Implemented in CrimeVision AI:

Feature ID & Name	Target User & Need	Technical Realization & Model Used	Implementation Status
FF-01: Multimodal Evidence Ingestion	Forensic Investigator uploading photos, audio, CCTV.	Drag-and-drop web panel supporting JPG, PNG, WAV, MP3, MP4 with file status validation.	**100% Implemented**
FF-02: RT-DETR Object & Weapon Detection	Detecting weapons, humans, and physical evidence.	Deep learning detection via PekingU/rtdetr_r50vd with normalized bounding boxes [xmin, ymin, xmax, ymax].	**100% Implemented**
FF-03: 3D Spatial Scene Reconstruction	Overcoming flat 2D photograph limitations.	Depth map estimation [X, Y, Z] projected into interactive Three.js WebGL canvas with OrbitControls.	**100% Implemented**
FF-04: Whisper Speech Transcription	Transcribing 911 calls & witness testimonies.	OpenAI Whisper-base automatic speech recognition pipeline with timestamp extraction.	**100% Implemented**
FF-05: 4-Language Translation Engine	Bridging language barriers during golden hour.	Groq LLM cross-lingual translation into English, Hindi, Telugu, and French tabs in UI.	**100% Implemented**
FF-06: OpenCV Video Evidence Engine	Surveillance footage analysis for suspects & vehicles.	OpenCV frame differencing and contour detection at 1.0 Hz for people, vehicles, weapons, 2D boxes, and 3D spatial trajectories.	**100% Implemented**
FF-07: Resilient Context-Aware Forensic Chatbot	Natural language evidence interrogation.	Dual-tier conversational agent (Groq LLaMA-3.3-70B + Local Forensic Reasoning Engine) answering people, vehicles, weapons, exact timestamps, with interactive UI chips.	**100% Implemented**
FF-08: Cross-Modal Multimodal Fusion	Reconciling discrepancies between modalities.	Unified temporal clock aligning video timestamps with audio facts and visual objects.	**100% Implemented**
FF-09: Court-Admissible PDF Generation	Submitting evidence dossiers to magistrates.	ReportLab Platypus synthesis with metadata, catalog tables, timeline, and SHA-256 verification hash.	**100% Implemented**
FF-10: Resilient Dual-Tier Persistence	Preserving evidence across sessions and reloads.	In-memory caching backed by JSON disk storage in backend/data/sessions/ with UUID keys.	**100% Implemented**
FF-11: Dedicated Audio Player & Waveform Equalizer	Acoustic forensics & audio playback.	Custom HTML5 audio transport controller, seek scrub slider, speed toggle, and animated 36-bar acoustic equalizer visualizer.	**100% Implemented**

Summary: Total 10 Major Functional Features fully architected, coded, tested, and validated in the CrimeVision AI platform.